

PROFESSIONAL SUMMARY

- M.Tech VLSI Design graduate with hands-on experience in Analog Layout, Custom IC Design, RTL Design and ASIC Physical Design, supported by academic projects across analog and digital design flows.
- Skilled in Cadence Virtuoso for schematic design, simulation, custom layout design, parasitic-aware layout optimization, and Physical Verification using Calibre and Assura DRC/LVS.
- Familiar with ASIC implementation flow including RTL design, synthesis, floorplanning, placement, CTS, routing, STA, and timing analysis using Synopsys tools.
- Worked on projects involving Low Noise Amplifier (LNA) design in SCL 180nm, low-power arithmetic circuits in GPDK 90nm, and RISC-V based ASIC implementation using SAED32nm technology.
- Familiar with common-centroid matching techniques, parasitic-aware layout optimization, and analog custom layout methodologies.

TECHNICAL SKILLS

HDLs: Verilog, SystemVerilog, Tcl

Digital / ASIC: RTL Design, FSM Design, Functional Verification, Synthesis, STA, Floorplanning, Placement, CTS, Routing

Analog / RF: Cadence Virtuoso, Analog/RF Simulation, S-parameters (S11, S12, S21, S22), Noise Figure (NF), Linearity (IIP3, OIP3, IMD3, P1dB), Stability (K-factor, μ -factor)

Layout Design: Custom Layout Design, Device Matching, Common-Centroid, Interdigitation, Guard Rings, Shielding, Manual Routing, Layout Optimization

Physical Verification: Calibre DRC/LVS, Assura DRC/LVS, Parasitic Extraction (PEX)

EDA Tools: Cadence Virtuoso, Synopsys VCS, Design Compiler, ICC2, Cadence Innovus, ModelSim, Quartus Prime, Yosys, Verilator, GTKWave

Process Technologies / PDKs: SAED 32nm, TSMC 65nm (Academic), GlobalFoundries 40nm, SCL 180nm, GPDK 180nm, GPDK 90nm, GPDK 45nm

Programming: C, Python, Perl

Others: Timing Optimization, Power Analysis, DRC/LVS Debugging

PROJECTS

Performance-Balanced Noise-Canceling Low Noise Amplifier

Apr 2026

Key Responsibilities:

- Designed and analyzed CG-CS, cascode, and folded-cascode LNA architectures in SCL 180nm technology using Cadence Virtuoso.
- Performed RF analysis including Gain, Noise Figure, S-parameters, Linearity, and Stability analysis.
- Implemented custom layout using common-centroid matching techniques with parasitic-aware optimization.
- Performed Calibre/Assura DRC, LVS, post-layout RC parasitic extraction (PEX), and layout verification following foundry design rules.

Ultra-Lightweight RISC-V Architecture for Anomaly Detection

Nov 2025

Key Responsibilities:

- Designed custom RISC-V RTL modules in Verilog and performed functional verification using Synopsys VCS.
- Synthesized the design using Design Compiler and analyzed timing, area and power optimization using SAED32nm technology libraries.
- Executed floorplanning, placement, CTS, routing, and STA using Synopsys ICC2.

Low Power Hybrid Full Adder for Array Multiplier

Mar 2025

Key Responsibilities:

- Designed 4-bit and 8-bit hybrid full-adder based array multipliers in GPDK 90nm technology using Cadence Virtuoso.
- Developed schematic and custom layout with layout optimization for low-power operation.
- Performed Physical Verification including DRC/LVS and post-layout parasitic extraction (PEX).

Memristor-Based Kogge–Stone Adder

Nov 2024

Key Responsibilities:

- Designed and simulated 4-bit and 8-bit memristor-based Kogge–Stone adders using Cadence Virtuoso.
- Evaluated delay and power performance against conventional CMOS implementations.
- Performed circuit simulation and architectural analysis for high-speed low-power applications.

QUALIFICATION

Master of Technology (M.Tech) in VLSI Design
Vellore Institute of Technology, Chennai 2024 – 2026

Bachelor of Engineering (B.E.) in Electronics and Communication Engineering
Sri Sairam Engineering College, Chennai 2019 – 2023

CERTIFICATIONS

NPTEL:

- [System Design Through Verilog](#)
- [VLSI Design Flow: RTL to GDS](#)